

Lab: Clinical Translation and Computational Psychiatry

Learning Goal: Apply Active Inference models to clinical disorders, design translational studies, and evaluate therapeutic implications.

Part 1: Disorder Modeling

Exercise: For each disorder, identify the primary computational dysfunction and design a behavioral task to measure it:

Disorder	Primary Dysfunction	Predicted Behavioral Pattern	Proposed Task
Schizophrenia	Aberrant sensory precision (too high)	Over-reliance on sensory input, resistance to perceptual priors	Force-fusion task: present ambiguous stimuli with varying prior probability — patients should be less influenced by manipulated priors
Depression	Pessimistic priors + reduced reward PE precision	Win-stay reduced, loss-shift increased, negative expectation bias	Probabilistic reward task with positive/negative feedback — patients should show blunted learning from positive feedback
GAD	Elevated threat PE precision	Hypervigilance, excessive threat detection, safety signal insensitivity	Threat/safety conditioning with reversal — patients should show persistent threat response despite safety signals
Autism	High sensory precision, low social precision	Excellent perceptual discrimination, difficulty with social inference	Visual search (high accuracy expected in ASD) + social inference task (reduced accuracy expected)

For one disorder of your choice, expand the model: specify all parameter values and generate predictions for 200 simulated trials.

Write your response here

Part 2: Translational Pipeline

Learning Goal: Design a complete translational study.

Exercise: Take the schizophrenia model through all 5 pipeline steps:

Step 1 — Computational theory: Schizophrenia involves aberrant precision in the sensory hierarchy. Specifically, NMDA receptor hypofunction reduces the precision of top-down predictions, making sensory prediction errors disproportionately influential.

Step 2 — Behavioral validation: Design a task (e.g., hollow mask illusion — do patients fail to see the illusion because their strong sensory PEs override the top-down "face" prior?)

Step 3 — Neural validation: Measure MMN in patients — reduced MMN suggests impaired prediction error generation (or is it increased PE due to aberrant precision? The direction of the prediction matters).

Step 4 — Pharmacological validation: Administer ketamine (NMDA antagonist) to healthy controls. Does it reproduce the computational profile seen in patients?

Step 5 — Treatment: If the dysfunction is NMDA-related, would NMDA agonists (e.g., glycine site agonists) restore normal precision? What computational parameters would you monitor?

Write your response here

Part 3: Clinical Trial Design

Learning Goal: Design a Bayesian adaptive clinical trial.

Exercise: Design a trial for a novel anxiety treatment targeting threat precision:

1. **Primary outcome:** Reduction in threat PE precision (measured computationally from threat/safety task)
2. **Secondary outcome:** Clinical symptom reduction (GAD-7 scale)
3. **Enrichment:** Only enroll patients with computational phenotype showing elevated threat precision (top quartile on baseline computational assessment)
4. **Adaptive design:** After every 10 patients, update the posterior on treatment effect. If $\Pr(\text{treatment effective}) > 0.95$, stop for efficacy. If $\Pr(\text{treatment effective}) < 0.05$, stop for futility.
5. **Sample size:** Estimate needed N for 80% power

What advantages does this computational enrichment + adaptive design offer over standard parallel-group RCT?

Write your response here

Part 4: Digital Phenotyping Scenario

Learning Goal: Design a digital phenotyping system.

Exercise: Design a smartphone-based monitoring system for depression:

1. **Daily task:** 2-minute probabilistic learning task on smartphone (takes 50 trials)
2. **Parameters tracked:** Learning rate (α), reward precision (ω_{reward}), loss precision (ω_{loss}), policy precision (β)
3. **Passive measures:** Activity level (accelerometer), social interaction (communication app usage), sleep (phone usage patterns)
4. **Predictive model:** Use changes in computational parameters to predict relapse 1-2 weeks before clinical symptoms emerge

Design the alert system: What parameter changes trigger a clinical alert? How do you balance sensitivity (catching relapses) vs. specificity (avoiding false alarms)?

Write your response here

Part 5: Reflection

In 300 words, reflect: Computational psychiatry promises to transform diagnosis and treatment by identifying precise computational dysfunctions. But psychiatric disorders are enormously complex — involving genetic, developmental, social, and environmental factors. Can a computational model capture enough of this complexity to be clinically useful? Where is the boundary between useful simplification and harmful reductionism?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Disorder modeling	Mapping dysfunction to parameters
2	Pipeline design	Translational validation
3	Clinical trial	Bayesian adaptive design
4	Digital monitoring	Real-time computational tracking
5	Critical evaluation	Reductionism vs. utility